Version 1.1

Physics PhD Atlas 2027

The International Applicant's Playbook (US, UK, Europe, Australia + Canada)

Version 1.1 · program data checked October 22, 2026 · 104 programs: 46 US, 12 UK, 25 rest of Europe, 2 Australia & NZ, 19 Canada

This playbook is the "how" that goes with the Atlas data. Every program-specific fact below comes from the Atlas, where each value links to an exact quote on the program's official page. General advice is labelled as advice. Nothing here is immigration, visa or legal advice. For those topics, use only official government sources (see Section 15).

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1. Start here (10 minutes)

1. Open **Atlas.html** in any browser. It works offline. Filter by what blocks you most (Physics GRE, English-test waiver, fee waivers), then star programs to build a shortlist.
2. Open **Atlas.xlsx** (Excel, Google Sheets, or LibreOffice). Pick your starred programs in the **Shortlist** tab. Deadlines, fees and rules fill in automatically. The **Budget**, **Recommenders** and **Supervisor outreach** tabs do the rest of the tracking.
3. Read Sections 2 to 9 of this playbook today. They cover the decisions with deadlines attached: fee waivers, English tests, GRE scores, the UK's funding competitions, Europe's position system, Australia's supervisor-first rules and Canada.
4. Before you pay any fee or book any test, click the value in the Atlas and open the official page. Programs change rules mid-cycle. When the Atlas says *Not stated*, the pages we checked did not say. It does **not** mean "no".

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2. Your remaining timeline (late-October start)

Treat these as working targets (advice). The fixed dates come from the programs.

When	Do this
This week	Shortlist 8 to 15 programs. Check which need the **Physics GRE** (Section 5) and whether your English-test plan covers every program (Section 4).

When	Do this
By **Nov 1, 2026**	If Rice is on your list, request its worldwide **Doctoral Application Fee Waiver Weeks** waiver; Rice says it then needs a completed application by **Nov 27**.
Oct 31 – Nov 3	Europe's first wave. EPFL's autumn round closes **Oct 31** (next round Apr 30). Heidelberg's HGSFP portal and IMPRS Quantum Dynamics close **Nov 1**. Imperial's first President's PhD Scholarship round closes **Nov 2**. IMPRS Astronomy (Heidelberg) closes **Nov 3**.
Early November	Email supervisors wherever they decide admission: Australia and New Zealand (Section 8), many UK and European programs (Sections 6–7) and Canada (Section 9). Send recommenders their packet (Section 12).
Nov 30 – Dec 3	IMPRS Quantum Materials (Dresden) closes **Nov 30** for its February selection workshop. The early US group closes too: Columbia, Duke, Northeastern, Ohio State, Arizona, CU Boulder, UT Austin, Vanderbilt. Several programs start reviewing the moment a deadline passes.
Mid-December	**Dec 15** is the most common US date: 24 of the 46 dated US programs here use it. Cambridge's funding deadline is **Dec 8**; Durham's is **Dec 15** for full scholarship consideration.
January	Most Canadian deadlines (Carleton, Concordia, Queen's, SFU, Calgary, Manitoba, Waterloo, York), plus a few US programs (see the table). Later or rolling in Canada: Dalhousie, McMaster, Guelph, Ottawa, Saskatchewan, Montréal, Western. Bristol's funding deadline: applications must arrive before **Jan 18**. Edinburgh's physics funding deadline is **Jan 22**.
Feb-April	Interviews and offers. Several programs in this Atlas (e.g., Stony Brook and Texas A&M) ask admitted students to respond by **April 15**. SISSA's spring call closes in early March; EPFL's second round on Apr 30.

Deadlines in this Atlas (earliest first)

Deadline	Program	Country	Note
Nov 1, 2026	Heidelberg (HGSFP)	Germany	year inferred; act by Nov 1, 2026
Nov 1, 2026	IMPRS Quantum Dynamics (Heidelberg)	Germany	act by Nov 1, 2026
Nov 2, 2026	Imperial College London	United Kingdom	act by Nov 2, 2026
Nov 3, 2026	IMPRS Astronomy (Heidelberg)	Germany	act by Nov 3, 2026
Nov 30, 2026	IMPRS Quantum Materials (Dresden)	Germany	act by Nov 30, 2026
Nov 30, 2026	Ohio State	USA	year inferred
Dec 1, 2026	Duke	USA	
Dec 1, 2026	Northeastern	USA	year inferred; sources conflict
Dec 1, 2026	Alberta	Canada	year inferred
Dec 1, 2026	Arizona	USA	
Dec 1, 2026	UBC	Canada	
Dec 1, 2026	CU Boulder	USA	year inferred

Deadline	Program	Country	Note
Dec 1, 2026	UT Austin	USA	year inferred
Dec 1, 2026	Vanderbilt	USA	
Dec 3, 2026	Columbia	USA	
Dec 7, 2026	Warwick	United Kingdom	act by Nov 16, 2026
Dec 8, 2026	Cambridge	United Kingdom	
Dec 8, 2026	Penn (UPenn)	USA	
Dec 11, 2026	Maryland	USA	
Dec 11, 2026	Toronto	Canada	year inferred
Dec 14, 2026	Stanford	USA	
Dec 14, 2026	UChicago	USA	
Dec 15, 2026	Boston University	USA	year inferred
Dec 15, 2026	Caltech	USA	year inferred
Dec 15, 2026	Cornell	USA	year inferred
Dec 15, 2026	Durham	United Kingdom	
Dec 15, 2026	Georgia Tech	USA	year inferred
Dec 15, 2026	Harvard	USA	
Dec 15, 2026	Indiana (Bloomington)	USA	year inferred
Dec 15, 2026	Johns Hopkins	USA	year inferred
Dec 15, 2026	MIT	USA	year inferred
Dec 15, 2026	McGill	Canada	year inferred
Dec 15, 2026	Michigan State	USA	
Dec 15, 2026	Purdue	USA	year inferred
Dec 15, 2026	Rice	USA	
Dec 15, 2026	Stony Brook	USA	year inferred
Dec 15, 2026	Berkeley	USA	
Dec 15, 2026	UC Irvine	USA	year inferred
Dec 15, 2026	UCLA	USA	
Dec 15, 2026	UMass Amherst	USA	year inferred
Dec 15, 2026	Minnesota	USA	year inferred
Dec 15, 2026	Notre Dame	USA	year inferred

Deadline	Program	Country	Note
Dec 15, 2026	Pittsburgh	USA	
Dec 15, 2026	Rochester	USA	year inferred
Dec 15, 2026	Virginia (UVA)	USA	year inferred
Dec 15, 2026	U Washington (Seattle)	USA	
Dec 15, 2026	UW-Madison	USA	
Dec 15, 2026	Yale	USA	
Dec 16, 2026	UCSB	USA	
Dec 18, 2026	Utah	USA	year inferred
Jan 1, 2027	Rutgers	USA	year inferred
Jan 4, 2027	NC State	USA	
Jan 4, 2027	St Andrews	United Kingdom	year inferred
Jan 7, 2027	Queen's	Canada	year inferred; rolling
Jan 8, 2027	Manitoba	Canada	year inferred
Jan 9, 2027	Brown	USA	year inferred
Jan 9, 2027	Waterloo	Canada	year inferred
Jan 10, 2027	Texas A&M	USA	
Jan 15, 2027	Arizona State	USA	
Jan 15, 2027	Carleton	Canada	year inferred
Jan 15, 2027	Concordia	Canada	year inferred
Jan 15, 2027	Penn State	USA	year inferred
Jan 15, 2027	SFU	Canada	year inferred; rolling
Jan 15, 2027	Buffalo	USA	year inferred
Jan 15, 2027	Calgary	Canada	year inferred
Jan 15, 2027	UIUC	USA	
Jan 15, 2027	York	Canada	
Jan 17, 2027	Bristol	United Kingdom	
Jan 22, 2027	Edinburgh	United Kingdom	
Jan 28, 2027	Queen Mary (QMUL)	United Kingdom	year inferred
Jan 31, 2027	Glasgow	United Kingdom	year inferred

Deadline	Program	Country	Note
Feb 1, 2027	McMaster	Canada	year inferred; rolling
Feb 1, 2027	Guelph	Canada	year inferred; sources conflict
Feb 1, 2027	Montréal	Canada	
Mar 1, 2027	Western	Canada	
Apr 1, 2027	Dalhousie	Canada	year inferred
Apr 15, 2027	ANU	Australia	year inferred
Apr 30, 2027	EPFL	Switzerland	year inferred
Not stated	Munich (MCQST)	Germany	
Not stated	SISSA	Italy	
Not stated	Melbourne	Australia	
Not stated	Ottawa	Canada	
Not stated	Padova	Italy	
Not stated	Paris-Saclay	France	
Per position	Chalmers	Sweden	
Per position	TU Delft	Netherlands	
Per position	ETH Zurich	Switzerland	
Per position	ICFO (Barcelona)	Spain	
Per position	KTH Stockholm	Sweden	
Per position	King's College London	United Kingdom	
Per position	Leiden	Netherlands	
Per position	Stockholm	Sweden	
Per position	TU Munich	Germany	
Per position	Amsterdam (UvA)	Netherlands	
Per position	Copenhagen (NBI)	Denmark	
Per position	Strathclyde	United Kingdom	
Per position	Zurich (UZH)	Switzerland	
Per position	Uppsala	Sweden	
Rolling	Aalto	Finland	
Rolling	IMPRS Solar System (Göttingen)	Germany	2026/27 round dates not yet published
Rolling	Royal Holloway	United Kingdom	

Deadline	Program	Country	Note
Rolling	Trinity College Dublin	Ireland	
Rolling	Bonn-Cologne (BCGS)	Germany	
Rolling	Saskatchewan	Canada	

"Year inferred" means the page gave a day and month but no year; we assumed the Fall 2027 cycle. Confirm on the page.

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3. Build a balanced list, and a budget

Advice. Aim for 8 to 15 applications. Choose by research fit first and prestige second. A program where two or more faculty work on what you want is a stronger application than a famous name where nobody does.

- **Fit test:** for each program, name two faculty members and one recent paper from each that you could discuss. If you can't, it isn't a fit yet.
- **Spread:** mix programs where your profile is typical of admits with a few reaches. Use each department's own descriptions (several state typical GPAs or class sizes on their pages) rather than forum rumours.
- **Cost reality:** the median international application fee in this Atlas is **90 USD (about ₹8,500)** for US programs and **135 CAD (about ₹9,100)** for Canadian programs, where stated. Ten US applications therefore cost roughly **₹85,000** before test-score fees. Many UK, European and Australian programs charge no application fee (Sections 6–8). Rupee figures use the Bank of Canada rate of October 21, 2026 (1 USD ≈ ₹94.5), and your bank adds a margin.
- **Budget it** in the **Budget** tab, which converts every currency to rupees with an editable rate. Add score-sending and transcript costs from the official sites; we don't estimate those for you.

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4. English tests in 2026: what changed and who waives

Verified change: TOEFL iBT moved to a new **1-6 score scale on January 21, 2026**. Programs now list two minimums, one per scale (e.g., UBC: 90 on the old scale or 4.5 on the new one). Check which scale your score is on.

There are four kinds of waiver rule. Which kind a program uses decides whether an English-medium Indian degree is enough:

1. **Friendly to an English-medium Indian degree ("Yes" in the Atlas):** Aalto, ANU, TU Delft, EPFL, Harvard, IMPRS Astronomy (Heidelberg), IMPRS Solar System (Göttingen), McGill, Northeastern, Rutgers, Stanford, Stockholm, Stony Brook, TU Munich, Arizona, Bristol, UBC, CU Boulder, Copenhagen (NBI), Notre Dame, UT Austin, Toronto. Most of these look at your university's language of instruction. Some ask for an official letter or transcript notation (e.g., Toronto), and some for documents sent to the department (McGill). UT Austin is different: its exempt-country list includes India itself.

2. **Conditional / list-based.** Exemption only if your institution meets a specific test. Alberta publishes a list that includes the IITs (all campuses), IISc and the University of Hyderabad but **not** the IISERs. Waterloo accepts only institutions listed on WHED with English as the *only* language of instruction, and explicitly rejects medium-of-instruction letters. Queen's lets you *request* an exemption after a recent year of English-medium study. In this Atlas: Arizona State, Brown, Chalmers, Columbia, Dalhousie, Duke, KTH Stockholm, IMPRS Quantum Materials (Dresden), McMaster, Queen Mary (QMUL), Queen's, Royal Holloway, Trinity College Dublin, Alberta, UC Irvine, Edinburgh, Guelph, Penn (UPenn), Saskatchewan, St Andrews, Virginia (UVA), U Washington (Seattle), Waterloo, Vanderbilt, Yale, York.
3. **No waiver for an English-medium Indian degree ("No"):** Carleton, Concordia, Durham, Georgia Tech, Imperial College London, Indiana (Bloomington), King's College London, MIT, Michigan State, SFU, Ohio State, Buffalo, Calgary, Berkeley, Cambridge, Glasgow, UIUC, Manitoba, UMass Amherst, Rochester, Utah, Warwick, Western. These either use an "English-primary country" list that India is not on (e.g., UIUC, Berkeley, SFU, Calgary, and Cambridge via the UK government's list), or require a test from everyone (Georgia Tech).
4. **Not stated** on the pages checked: ask the program before you assume.

Advice: if even one program on your list needs a score, book the test now. Score delivery takes time; Waterloo, for example, says it takes 4-6 weeks for official scores to reach it. Also check speaking minimums. Several programs tie teaching-assistant eligibility (and so part of your funding) to a separate, higher speaking score.

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5. GRE and Physics GRE: required again at some top programs

Most programs in this Atlas don't need the GRE, and several will **not even look** at it. A notable group requires the **Physics GRE** for this cycle:

- **Brown**: Required for the Physics track; not considered for the Astrophysics track; waived if you hold a master's in physics or a related field
- **Caltech**: Physics GRE REQUIRED for the Fall 2027 cohort
- **Cornell**: REQUIRED for theory applicants (theoretical astro, particle, condensed matter); theory applications without it cannot be considered
- **Harvard**: Physics GRE subject test required
- **Stanford**: Physics GRE REQUIRED for all applicants
- **Calgary**: REQUIRED for all foreign applicants
- **Berkeley**: Physics GRE REQUIRED for Fall 2027 entry
- **UChicago**: Physics GRE REQUIRED for the 2026-2027 cycle
- **Guelph**: Physics GRE required for applicants whose degrees are not from a Canadian institution (minimum 800); an application below a minimum is still considered as a whole
- **Virginia (UVA)**: Valid Physics GRE score required (self-reported at application; official scores needed on matriculation)

Programs that will not consider **any** GRE score, even if you send one: Rice, Stony Brook, UC Irvine, UCSB, Minnesota, Notre Dame, Vanderbilt. Programs that ignore the **General** GRE but still review the Physics GRE: UT Austin, UW-Madison.

Advice:

- If your list contains any program in the first group, you need a Physics GRE score that arrives **before that program's deadline**. Check the remaining ETS test dates and score-reporting times on ets.org today; don't rely on forum dates.
- If you have a strong score and a program marks it *Optional* or *Recommended*, sending it can help. UT Austin is a special case: it asks you to explain in your Statement of Purpose why your Physics GRE matters, or it ignores the score.
- If you have no score and your list has none of the "Required" programs, don't spend scarce weeks on it now.

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6. The UK: funding competitions, and English rules for Indian degrees

The 12 UK programs in this Atlas mostly work like this:

- you apply to a department, usually naming a project or a supervisor;
- funding is decided in separate studentship and scholarship competitions, which close well before the final admission deadline.

Program	Funding deadline	Fee	Supervisor first?	English route for an Indian degree	Funding	Entry
Imperial College London	Nov 2, 2026	0 (no fee)	Not required	No	Competitive	Direct from bachelor's
Warwick	Dec 7, 2026	0 (no fee)	Recommended	No	Competitive	Direct from bachelor's
Cambridge	Dec 8, 2026	20 GBP	Recommended	No	Competitive	Master's usually required
Durham	Dec 15, 2026	Not stated	Recommended	No	Competitive	Either
St Andrews	Jan 4, 2027	Not stated	Recommended	Conditional	Competitive	Direct from bachelor's
Bristol	Jan 17, 2027	Not stated	Recommended	Yes	Competitive	Master's usually required
Edinburgh	Jan 22, 2027	Not stated	Not required	Conditional	Competitive	Direct from bachelor's
Queen Mary (QMUL)	Jan 28, 2027	0 (no fee)	Required	Conditional	Competitive	Direct from bachelor's
Glasgow	Jan 31, 2027	0 (no fee)	Required	No	Competitive	Direct from bachelor's
King's College London	Per position	Not stated	Required	No	Competitive	Direct from bachelor's
Royal Holloway	Rolling	Not stated	Recommended	Conditional	Competitive	Master's usually required

Program	Funding deadline	Fee	Supervisor first?	English route for an Indian degree	Funding	Entry
Strathclyde	Per position	Not stated	Not required	Not stated	Competitive	Either

Apply by the funding deadline, not the final one. Examples:

- Imperial's President's PhD Scholarship rounds: Nov 2, Jan 11 and Mar 1.
- Warwick's Chancellor's Scholarship: Dec 7.
- Cambridge: Dec 8.
- Durham: Dec 15.
- Bristol: before Jan 18.
- Edinburgh: Jan 22.
- QMUL studentships: Jan 28.
- Glasgow studentships: Jan 31.

Funding for international students is limited.

- St Andrews says research councils permit only 30% of their funded places to go to international applicants.
- Warwick says its funded projects are in most cases limited to UK nationals.
- Imperial says studentships for overseas applicants are very restricted.

Without funding, international tuition is high: Glasgow lists £33,708 a year for 2027/28. Apply for every scholarship you're eligible for, and check each university's India page for country-specific schemes (advice).

No application fee at many UK universities: Imperial College London, Queen Mary (QMUL), Glasgow, Warwick. Cambridge charges £20 per applicant per year for up to 7 doctoral applications.

English is the biggest UK trap for Indian applicants.

- **No route without a test.** Most UK universities exempt only degrees from UKVI "majority English-speaking" countries, and India isn't one of them. In this Atlas, no exemption route for an Indian degree: Durham, Imperial College London, King's College London, Cambridge, Glasgow, Warwick.

- **India-specific evidence accepted elsewhere:**

    - QMUL accepts at least 70% in the English component of CBSE or ISC Class XII.
    - Royal Holloway accepts Class 12 English of at least 80% for research degrees, or a degree from a listed Indian university.
    - Bristol and St Andrews accept an official medium-of-instruction letter.
    - Edinburgh accepts English-taught degrees only from institutions on its approved list.

    In this Atlas, some India route exists: Queen Mary (QMUL), Royal Holloway, Bristol, Edinburgh, St Andrews.

- **Advice:** if your list mixes both kinds, book IELTS or TOEFL anyway. It's the only evidence every UK university accepts.

A strong bachelor's is often enough. Most UK programs ask for a first or 2:1 honours degree or the international equivalent: Durham, Imperial College London, King's College London, Queen Mary (QMUL), Edinburgh, Glasgow, St Andrews, Strathclyde, Warwick. Check the university's India page, though: Royal Holloway's says Indian applicants are considered for a PhD after a master's.

Name a supervisor, and plan for ATAS.

- Glasgow, King's College London and QMUL want you to contact or name a supervisor before applying; others recommend it.
- King's College London notes that successful applicants whose nationality isn't exempt need an **ATAS** security clearance certificate, which takes time. Check the UK government's ATAS page as soon as you get an offer.

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7. The rest of Europe: positions, calls and master's-first rules

The 25 programs in continental Europe and Ireland (Denmark, Finland, France, Germany, Ireland, Italy, Netherlands, Spain, Sweden, Switzerland) use two very different ways in.

1. **Calls with a deadline.** You apply to a graduate school or program by a published date, often followed by interviews. Examples:
    - Nov 1: IMPRS Quantum Dynamics, and Heidelberg's HGSFP portal.
    - Nov 3: IMPRS Astronomy (Heidelberg).
    - Nov 30: IMPRS Quantum Materials (Dresden).
    - Spring: SISSA's exam-based call.
    - Oct 31 and Apr 30: EPFL's two rounds.
2. **Apply for a paid position, or agree a supervisor first.** There is no central committee: PhD places are advertised as jobs, or a professor must accept you before you apply. Sweden (Stockholm, KTH, Chalmers, Uppsala), Denmark (Copenhagen), the Netherlands (Leiden, Amsterdam, Delft), Switzerland (ETH Zurich, Zurich), TU Munich, ICFO and Trinity College Dublin work this way. In this Atlas, position- or supervisor-based: Aalto, Chalmers, TU Delft, EPFL, ETH Zurich, ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Solar System (Göttingen), Stockholm, TU Munich, Trinity College Dublin, Bonn-Cologne (BCGS), Amsterdam (UvA), Copenhagen (NBI), Zurich (UZH), Paris-Saclay, Uppsala.

Program	Country	Deadline	Supervisor first?	Funding	Entry	Fee
Copenhagen (NBI)	Denmark	Per position	Required	Guaranteed	Master's usually required	Not stated
Aalto	Finland	Rolling	Required	Competitive	Master's usually required	Not stated
Paris-Saclay	France	Not stated	Required	Guaranteed	Master's usually required	Not stated
Bonn-Cologne (BCGS)	Germany	Rolling	Required	Not stated	Not stated	Not stated
Heidelberg (HGSFP)	Germany	Nov 1, 2026	Not required	Not stated	Master's usually required	Not stated

Program	Country	Deadline	Supervisor first?	Funding	Entry	Fee
IMPRS Astronomy (Heidelberg)	Germany	Nov 3, 2026	Not required	Guaranteed	Master's usually required	Not stated
IMPRS Quantum Dynamics (Heidelberg)	Germany	Nov 1, 2026	Not stated	Competitive	Master's usually required	Not stated
IMPRS Quantum Materials (Dresden)	Germany	Nov 30, 2026	Not required	Guaranteed	Either	0 (no fee)
IMPRS Solar System (Göttingen)	Germany	Rolling	Not required	Typical	Master's usually required	Not stated
Munich (MCQST)	Germany	Not stated	Not stated	Not stated	Not stated	Not stated
TU Munich	Germany	Per position	Required	Typical	Master's usually required	Not stated
Trinity College Dublin	Ireland	Rolling	Required	Typical	Direct from bachelor's	Not stated
Padova	Italy	Not stated	Not stated	Competitive	Master's usually required	Not stated
SISSA	Italy	Not stated	Not stated	Not stated	Not stated	Not stated
Amsterdam (UvA)	Netherlands	Per position	Required	Typical	Master's usually required	Not stated
Leiden	Netherlands	Per position	Required	Guaranteed	Master's usually required	Not stated
TU Delft	Netherlands	Per position	Required	Typical	Master's usually required	Not stated
ICFO (Barcelona)	Spain	Per position	Required	Guaranteed	Master's usually required	Not stated
Chalmers	Sweden	Per position	Not stated	Guaranteed	Master's usually required	0 (no fee)
KTH Stockholm	Sweden	Per position	Required	Guaranteed	Master's usually required	Not stated
Stockholm	Sweden	Per position	Required	Guaranteed	Master's usually required	Not stated
Uppsala	Sweden	Per position	Required	Guaranteed	Master's usually required	0 (no fee)
EPFL	Switzerland	Apr 30, 2027	Required	Guaranteed	Master's usually required	0 (no fee)
ETH Zurich	Switzerland	Per position	Required	Typical	Not stated	Not stated
Zurich (UZH)	Switzerland	Per position	Required	Typical	Master's usually required	100 CHF

Fees are often low or zero.

- **No application fee:** Chalmers, EPFL, IMPRS Quantum Materials (Dresden), Uppsala.
- **Small fees:** Zurich (UZH) charges CHF 100.
- **Watch for tuition instead.** From January 2027, Copenhagen charges PhD tuition of DKK 85,000 a year, which your funding must cover.

A master's is usually expected first. In this Atlas: Aalto, Chalmers, TU Delft, EPFL, Heidelberg (HGSFP), IMPRS Astronomy (Heidelberg), ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Quantum Dynamics (Heidelberg), IMPRS Solar System (Göttingen), Stockholm, TU Munich, Amsterdam (UvA), Copenhagen (NBI), Padova, Zurich (UZH), Paris-Saclay, Uppsala. Bachelor's holders can enter at: IMPRS Quantum Materials (Dresden), Trinity College Dublin. Other routes:

- Heidelberg (HGSFP and IMPRS Astronomy) has 4+4 tracks for excellent candidates with a 4-year bachelor's.
- EPFL accepts 4- or 5-year bachelor's holders who explain their education in the Statement of Objectives.
- Swedish rules (KTH, Stockholm) accept 240 credits including at least 60 at advanced level, so a 4-year degree with enough advanced courses can qualify.

If you hold a 5-year BS-MS, ask each program how it assesses your degree, and keep the reply.

Funding usually means a job contract.

- **Sweden:** funding for the whole doctorate is a condition of admission, and doctoral students are usually salaried. KTH quotes SEK 33,000–37,100 a month.
- **Funded-only admission elsewhere:** Copenhagen, ICFO and Paris-Saclay admit only students who are funded.
- **Positions:** Leiden's PhD projects are 4-year paid positions, and ETH doctorates are, with few exceptions, linked to a position.

In this Atlas, paid or typically funded: Chalmers, TU Delft, EPFL, ETH Zurich, IMPRS Astronomy (Heidelberg), ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Quantum Materials (Dresden), IMPRS Solar System (Göttingen), Stockholm, TU Munich, Trinity College Dublin, Amsterdam (UvA), Copenhagen (NBI), Zurich (UZH), Paris-Saclay, Uppsala; competitive: Aalto, IMPRS Quantum Dynamics (Heidelberg), Padova.

English rules for Indian degrees are often friendlier than in the UK.

- **Friendly:**
  - Aalto lists India for English-medium degrees.
  - Stockholm and IMPRS Solar System count an English-taught degree.
  - TU Delft exempts English-taught master's degrees in principle.
  - IMPRS Astronomy (Heidelberg) and Copenhagen assess English at interview instead of by test.
- **School-level English counts in Sweden:** at KTH and Chalmers, your CBSE or CISCE Class XII English grades can meet the requirement.

Advice.

- In position-based systems, your supervisor email (Section 10) *is* the application. Check the vacancy pages weekly; KTH, for example, announces positions nine times a year.
- Allow time after an offer for contracts and permits. EPFL says about 3 months is usually needed for visas and work permits.

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8. Australia and New Zealand: supervisor first, scholarship rounds

The 2 programs in this Atlas (Australia) share a pattern.

Program	Country	Next scholarship deadline	Fee	Supervisor first?	English route	Funding	Entry
ANU	Australia	Apr 15, 2027	0 (no fee)	Required	Yes	Competitive	Either
Melbourne	Australia	Not stated	0 (no fee)	Required	Not stated	Competitive	Either

- **Find a supervisor before you apply.** ANU requires written approval from an identified supervisor, and Melbourne needs a prospective supervisor's written support before you apply. In this Atlas, supervisor required: ANU, Melbourne.
- **No application fee for research degrees** at ANU, Melbourne.
- **Scholarships run in rounds and are competitive.** ANU says an offer of admission doesn't guarantee a scholarship. Its main international round for starts by March 2027 closed on Aug 31, 2026, and later rounds are shown in the table. The Australian academic year starts around February–March.
- **Entry needs research experience.**
  - ANU asks for an honours bachelor's of at least Second Class Upper (H2A), or equivalent research.
  - Melbourne asks for a four-year bachelor's whose research component is at least 25% of a full-time year, with a final-year average of 75%, or a master's.
  - Advice: a 4-year BS or a BS-MS with a thesis fits these rules better than a 3-year BSc. Ask the university if you're unsure.
- **English:** ANU accepts a bachelor's of at least 3 years taught and assessed entirely in English. In this Atlas, an English-medium route exists at: ANU.

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9. Canada: supervisors and master's-first rules

Supervisor first

In Canada, admission often depends on a professor agreeing to take you. In this Atlas:

Program	Supervisor before admission	What the page says
Carleton	Not stated	Not stated on the page checked

Program	Supervisor before admission	What the page says
Concordia	Required	No admission offer until a supervisor match is made; name potential supervisors in the statement of purpose and contact them (the department may help match)
Dalhousie	Required	Admission depends on finding a supervisor through the free pre-application/interview; the formal application is submitted only with a supervisor's support and names the supervisor
McGill	Recommended	Not required, but highly recommended: professors select students for their groups
McMaster	Recommended	Funding includes a research fellowship from the supervisor's grant; contact faculty directly to ask if they are taking students
Queen's	Recommended	A complete application needs a research proposal; work with a potential supervisor early
SFU	Not stated	Not stated in the text checked; transfers need supervisory committee approval
Alberta	Not stated	Physics pages do not state a pre-application supervisor requirement; FGPS notes some thesis programs require one
UBC	Not required	Name faculty of interest in the application; no prior commitment needed, but contacting faculty is encouraged
Calgary	Required	No supervisor needed to apply, but you are admitted only if at least one potential supervisor will fund you; contacting faculty first helps
Guelph	Required	You do not need a supervisor before applying (list 2-3 potential supervisors), but approved applicants are admitted only if a supervisor match is found
Manitoba	Required	Department cannot admit without a formal advisor commitment; contact a prospective advisor (via the associate head) before applying, and name your preferred supervisor in the application
Ottawa	Required	Supervisor's name required AT APPLICATION; you must be accepted by a thesis supervisor to register
Saskatchewan	Recommended	Supervisor approval recommended before applying; applications without a confirmed supervisor are circulated to faculty
Toronto	Not required	No supervisor needed to apply; MSc and 5-year direct-entry PhD students pick one after arrival; 4-year PhD entrants (with a master's) must secure one before registration, typically by May
Waterloo	Required	Admission requires a faculty member willing to supervise you; contact before or after applying and name supervisors on the application
Montréal	Recommended	University guidance: for the doctorate, find a research supervisor preferably before admission, or at the latest by the end of the 2nd term (unless the department requires it at admission)
Western	Required	Every admission offer is made together with the student's faculty supervisor; contacting potential supervisors before applying is strongly encouraged

Program	Supervisor before admission	What the page says
York	Not stated	Not stated on the page checked

Master's-first rules

Many Canadian PhD programs expect a master's degree. A BSc holder usually starts in the MSc and may transfer to the PhD later. In this Atlas, "Master's usually required": Carleton, Concordia, McGill, Queen's, SFU, UBC, Guelph, Manitoba, Ottawa, Saskatchewan, Waterloo, Montréal, Western, York. Direct or either route: Dalhousie, Alberta, Toronto.

Important for 5-year integrated BS-MS students (IISER, IISc, IIT and others): whether your degree counts as a master's equivalent is decided by each university's credential assessment. **Ask each program in writing.** Don't assume either way. Put the question in your first email to the graduate office and keep the reply.

Funding is stated more often

Canadian programs often publish guaranteed minimums (e.g., UBC at least CAD 40,000 a year for years 1-4; Alberta CAD 100,000 over four years; McMaster at least CAD 21,000 after tuition and fees). Note whether a figure is before or after tuition.

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10. Supervisor emails that get answered (advice)

Find people, not just programs. Use faculty pages, the group's own website (look for "accepting students"), and papers from the last two years. Shortlist 2-4 people per university.

The 6-sentence email:

> **Subject:** Prospective PhD student (Fall 2027): [specific topic] in your group
>
> 1. Who you are: degree, institution, expected graduation.
> 2. The specific work of theirs you read, and one genuine question or connection.
> 3. Your most relevant research experience and what *you* did in it.
> 4. One skill or result that would help their group (a technique, code, measurement).
> 5. The ask: "Are you planning to take new PhD students for Fall 2027, and would you be open to my application?"
> 6. Thanks, with CV and transcript attached as PDFs named Lastname_CV.pdf.

Cadence: one polite follow-up after about 10 days if you get no reply, then move on. Log every email in the **Supervisor outreach** tab; it calculates the follow-up date for you.

Don'ts: no mass emails with swapped names, no long life stories, no attachments over a few MB, and no generic praise ("your esteemed research").

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11. Statement of purpose: a physics framework (advice)

Programs set their own length rules. Alberta, for example, wants **500 words or less on a single page**. Read each program's prompt and obey it exactly.

A structure that works for research PhDs:

1. **Opening (2-3 sentences):** the physics question that drives you, stated concretely ("how disorder changes spin dynamics in..."), not "I have loved physics since childhood".
2. **Research experience (40-50% of the text):** for each project, give the question, *your* contribution, the method, the result, and what you learned, including what went wrong and how you fixed it.
3. **Skills and preparation:** techniques, instruments, code, and advanced courses that map onto their work.
4. **Why this program:** two named faculty and why, plus facilities or centres that matter for your plans.
5. **Goals:** where the PhD leads and why this training fits.

Check before you submit: every sentence should be true and specific, and you should be able to defend it in an interview. Write it yourself. Never use purchased or ghost-written statements.

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12. Recommendation letters: the request kit (advice)

- **Who:** research supervisors first, then instructors of advanced courses who know your work. Programs differ on number: Toronto wants **exactly two** (no more), while UIUC, Columbia and Maryland list three.
- **When:** ask at least 4-6 weeks before your earliest deadline, and send portal invitations early.
- **What to send (one email, one PDF):** your shortlist with deadlines and portal links, CV, draft SOP, a "brag sheet" (projects, dates, what you did, skills shown), and a reminder of specific moments they saw your work.
- **Reminders:** a friendly note 10 days and again 3 days before each deadline. The **Recommenders** tab tracks status per program.

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13. Fee waivers, ethically

What this Atlas found for international applicants:

- **Free or waived for everyone this cycle:** Penn State. **No application fee at all:** ANU, Chalmers, EPFL, Imperial College London, IMPRS Quantum Materials (Dresden), Queen Mary (QMUL), Glasgow, Melbourne, Warwick, Uppsala.
- **Open to internationals (need, hardship, or a time-boxed window):** Columbia, Cornell, Rice, Notre Dame, Pittsburgh, Yale.
- **Limited or conditional routes:** Arizona State, Brown, Caltech, Duke, Georgia Tech, MIT, Purdue, Rutgers, Stanford, Ohio State, Arizona, UBC, UCLA, Cambridge, UChicago, CU Boulder, UMass Amherst, Penn (UPenn), Rochester, UT Austin, Utah, Virginia (UVA), Vanderbilt.

- **Domestic applicants only:** Boston University, Michigan State, NC State, Stony Brook, Texas A&M, Buffalo, Berkeley, UC Irvine, UCSB, U Washington (Seattle), UW-Madison.
- **No waivers:** Carleton, Concordia, Indiana (Bloomington), Trinity College Dublin, Alberta, Calgary, Toronto.

Advice:

- Request a waiver **before** paying. Application fees are typically non-refundable (UChicago's Physical Sciences Division says so for its $90 fee), so a fee paid too early is gone.
- Ask briefly and honestly: state your circumstances and the program you're applying to, and attach evidence if asked. Never misstate citizenship, income or event attendance.
- Some departments have small waiver budgets that run out months before the deadline (Rutgers says so), so ask early.

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14. Submission checklist

- ☐ Program-specific statement prompt answered (length, format, required sections)
- ☐ CV in PDF, with research section, publications/posters, skills
- ☐ Transcripts uploaded for every post-secondary institution (some programs, e.g., Toronto, ask for official copies only after admission)
- ☐ English score sent from the testing agency, or waiver evidence ready
- ☐ GRE / Physics GRE sent if required (check institution codes on each page)
- ☐ Recommenders invited, and each has confirmed they will submit
- ☐ Fee paid or waiver approved, with the receipt saved
- ☐ Supervisor names listed where the program asks (Waterloo, Ottawa, UBC and others)
- ☐ Final check of every value in the Atlas against the live official page

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15. Money, safety and what this kit does not cover

- **Scams:** be wary of anyone who guarantees admission, sells "special" fee-waiver codes, or asks for your portal passwords. Real waiver codes come from the program itself; Penn State, for example, publishes its own code on its admissions page.
- **Paying fees:** many portals take Visa or Mastercard credit cards. Some accept debit cards selected as "credit" (UBC says this). Enable international online transactions with your bank before the deadline week.
- **Visas and study permits:** out of scope. Use only official sources: travel.state.gov for US student visas, canada.ca (IRCC) for Canadian study permits, and for Europe the host university's international office and the national immigration authority it points you to. Paid immigration advice in Canada is a regulated activity, and this kit is not that.

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16. How this data was built (and how to report an error)

- Each of the 104 programs was checked against its own department and graduate-school pages on October 22, 2026. Every value stores the exact sentence it came from, the URL and the date. A script confirms that each quote appears verbatim on the fetched page.
- Research and drafting were AI-assisted. Every published value is tied to an official quote, and uncertain items are labelled *Not stated*, *Conditional*, *year inferred* or *sources conflict* instead of guessed.
- A weekly automated check re-reads the cited pages and flags quotes that disappear. Updates are sent to buyers as new versions at no charge.
- Some well-known programs couldn't be verified on the check date because their sites blocked automated checks or were down for maintenance. They are not in this edition: Oxford, UCL, Manchester, ISTA Austria, Vienna, Göttingen's central pages, Geneva, KU Leuven, ENS Paris (ICFP) and the University of Barcelona. They'll be added once their pages can be verified.
- **Found something wrong or changed?** Reply to your purchase email with the program and the page link. Confirmed corrections ship in the next update.